



Shell Petroleum Development Company of Nigeria

Classification Status: Restricted

Onshore NOV Team

Egbema Main D8000X, D9000X and E1000X Static Modeling and Volumetric Report

Document Number		Prepared by	Maryam Tafida (PG)
Document Owner	NOV Onshore Team	Contributors	Indutimi Preye (PG) Adedoyin Orekoya (PG) Francis Dike (RE) Ebimobowei Wodu (PP)

Agreed by:	
Preye, Indutimi UPO/G/DNN Define and Studies Lead, NOV Onshore Team	
Approved by:	
Alaka, Collins UPO/G/DNN Hub Dev. Manager, NOV Onshore Team	Ugboaja, Remmy UPO/G/DNC PG Discipline Principal (Onshore)
	Remmy Ugboaja Digitally signed by Remmy Ugboaja Date: 2017.09.26 17:13:36 +01'00'

Executive Summary

This report documents the static geological modeling of Egbema main D8000X, D9000X complex and E1000X Reservoirs and their respective hydrocarbon initially in-place (HCIIP) estimated.

Egbema field is located in SPDC's OML 20, about 100km NNW of Port Harcourt. The field was discovered in October 1960, and by mid-1967, seven wells had been drilled. The eighth well was drilled in September 1982, following a field review carried out in 1978. Another field review was done in 1987 and proposed a ninth well, which was drilled in 1992.

The primary objective of this Egbema D8000, D900 & E1000 static modelling is to validate NPDC volumes and optimise the development plan for the target reservoirs, as well as establish for SPDC an audit trail of Egbema D8000X, D9000X and E1000X reservoirs HCIIP volumes.

Volumetric computation and uncertainty analysis was carried out with average petrophysical properties (Porosity, Net-to-gross, and Saturation) as it was agreed to keep the model simple since it is pre-DG1.

Low and high case structure surfaces were created from base case to address the uncertainty in structure. Similarly, for the NiG, porosity, saturation and formation Volume factor the low and high case ranges were evaluated and used in sensitivity analysis.

The resulting tornado chart on the STOIP and GIIP volumes input parameters suggested 'Structure Depth' and Saturation as the key uncertainties having a major impact on the Egbema D8000, D9000 and E1000 reservoirs volumetrics. The tornado plots results were used for the second step which involves stringing of the various input parameters to generate the final technical limit low, base and high case HCIIP Volumes considering existing performance and planned development recoveries.

The ranges of volumes for the D8000, D9000 and E1000 are shown in the Table below.

Reservoir	STOIP (MMstb)			GIIP (Bcf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000	24.09	37.01	55.21	1.99	4.10	7.94
D9000	82.93	98.61	119.98	9.80	14.75	21.39
E1000	36.06	51.91	68.97	33.08	42.95	67.03

A comparison base case STOIP and GIIP for D8000, D9000 & E1000 with the ARPR 2017 are stated in the Table below.

	Current Studies		ARPR (01.01.2017)		Variation to ARPR 2017 (%)	
	STOIP	GIIP	STOIP	GIIP	STOIP	GIIP
D8000	37.01	4.10	37.01	4.10	0	0
D9000	98.61	14.75	84.80	16.70	-16	-12
E1000	51.91	42.95	60.80	40.20	15	-7

Table of Contents

Executive Summary	2
1. Introduction	4
1.1. Modelling Objectives	4
1.2. Data Availability	5
1.2.1. Wells	5
1.2.2. Seismic Data	5
2. Seismic Interpretation.....	6
2.1. Depth Uncertainty Estimation	8
3. Reservoir Geology	10
3.1. Structural Geology	10
3.2. Stratigraphy and Sedimentology	10
3.3. Conceptual Depositional Model	11
3.4. Correlation	11
4. Rock and Fluid Properties	12
4.1. Contacts	12
4.2. Rock Properties	13
4.3. PVT	14
5. Static Reservoir Modelling	16
5.1. Structural Modelling	16
5.2. Property Modelling	16
6. Uncertainty.....	17
7. Volumetric Summary.....	19
8. Conclusion & Recommendation.....	23
References.....	24
Appendix 1: Egbema D8000, D9000 & E1000 Top Structure Maps, cross section & correlation panel	25
Appendix 2: Egbema D8000, D9000 & E1000 Sand Tops And Bases Update	30

1. INTRODUCTION

Egbema field is located in SPDC's OML 20, about 100km NNW of Port Harcourt (Figure 1.1). The field was discovered in October 1960, and by mid-1967, seven wells had been drilled. The eighth well was drilled in September 1982, following a field review carried out in 1978. Another field review was done in 1987 and proposed a ninth well, which was drilled in 1992. The field comprises of 12 identified hydrocarbon reservoirs. It has a total STOIP of about 226 MMstb and a FGIP of about 302 Bscf. The developed and produced reservoirs in the field are D1000, D8000X, D9000X and E1000X. The field started production in April 1970, and achieved a peak of 18,000 bopd in 1972. Production declined steadily to 5500 bopd in 1982 when it was shut in due to offtake restrictions. The field was reopened in 1984 and has since been in decline, with current production of 1,056 bopd at 62% water cut and GOR of 700 scf/bbl from 3 drainage points on D1000X, D8000X, D9000X and E1000X Reservoirs. Cumulative production to date stands at 77 MMstb of oil and 56 Bscf of gas.

In 2010, the operatorship of the field along with other fields in Egbema Node was transferred to the Nigerian Petroleum Development Company (NPDC). NPDC is currently embarking on Egbema FOD study to identify further oil and gas development potentials in Egbema main field.

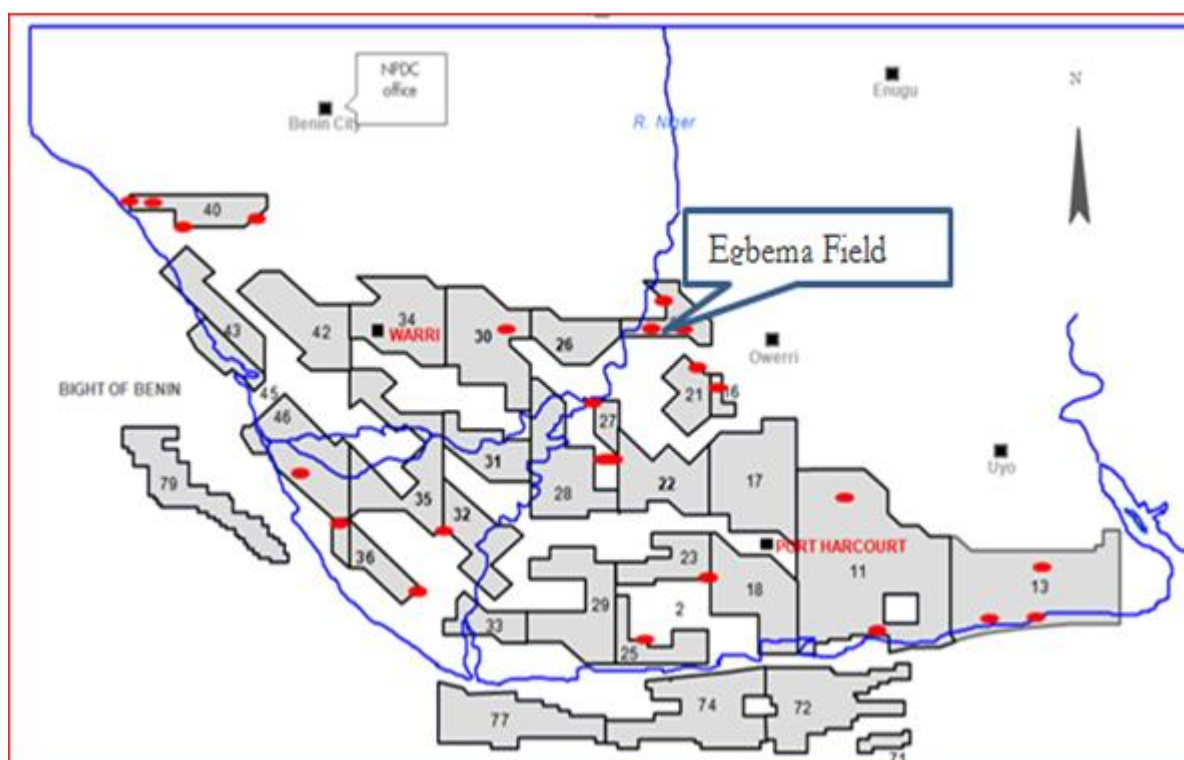


Figure 1-1: Location Map of Egbema Field

1.1. Modelling Objectives

The objectives of the modelling of Egbema D8000, D9000 and E1000 reservoirs are:

- Validate NPDC volumes and optimise the development plan for the target reservoirs

- Establish for SPDC an audit trail of Egbema D8000X, D9000X Complex and E1000X reservoirs HCIIP volumes
- Compare and update SPDC's NNS Volumes
- Provide input to 2017 URCR

1.2. Data Availability

1.2.1. Wells

A total of nine wells and a sidetrack are available over the Egbema Main Field. Table 1-1 shows the wells and available well logs. No core and pressure data were available in the field, Side wall samples were acquired in Egbema-01, Egbema -02 and Egbema -03.

Table 1-1: Data Availability table for Egbema Field

Well	GR			SP	CAL		RES			DENS			NEUT		SONIC	SWS
	GRN	GR	GR_008		CAL	CALI	SN	RT	MSFL	FDC	RHOB	DRHO	CNL	NEUT		
EGBM-001	No	No	No	Yes	Yes	No	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes
EGBM-002	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	No	Yes	No	No	No	No	Yes
EGBM-003	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes
EGBM-004	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No
EGBM-005	Yes	No	No	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No
EGBM-006	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes	No	Yes	No	No	No	No	No
EGBM-007	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes	No	Yes	No	No	No	No	No
EGBM-008	No	No	Yes	No	No	No	Yes	Yes	No	Yes	No	No	No	No	No	No
EGBM-009	Yes	No	No	No	No	Yes	Yes	Yes	No	No	No	Yes	No	No	Yes	No
EGBM-009-ST1	Yes	No	No	No	No	No	No	Yes	No	No	Yes	No	No	No	No	No

1.2.2. Seismic Data

3D seismic was acquired across OML 20 in 1994 and processed to post-stack time migration by SPDC SIPMAP,(1996) and a pre-stack time migration in 2009. Further re-processing of the 3D seismic was carried out in 2013 to post stack depth migration (PSDM) by IDSL which was used for this study. This re-processed (PSDM) seismic volume was transferred to SPDC for re-interpretation of the D8000X, D9000X Complex and E1000X reservoirs). NPDC provided only depth domain seismic volume, but a quick inspection showed that there is an acceptable well to seismic tie.

2. SEISMIC INTERPRETATION

SPDC re-interpreted the D8000, D9000 and E1000 horizons, and confirmed that the NPDC interpretation for D8000 and D9000 reservoirs was adequate (Figure 2-1). In the case of E1000X, reservoir, because of the poor lateral seismic continuity (seismic doublet) (Figure 2-2), three different reflectors were picked by SPDC at different depths; the deepest of these reflectors was interpreted to be the E1000X reservoir, because there is a better well to seismic tie with the E1000X well correlation (Figure 2-3). SPDC did not carry out a full field seismic interpretation of the Egbema main field. The SPDC interpretation of the candidate reservoirs provided the basis for the current field re-evaluation and development premises.

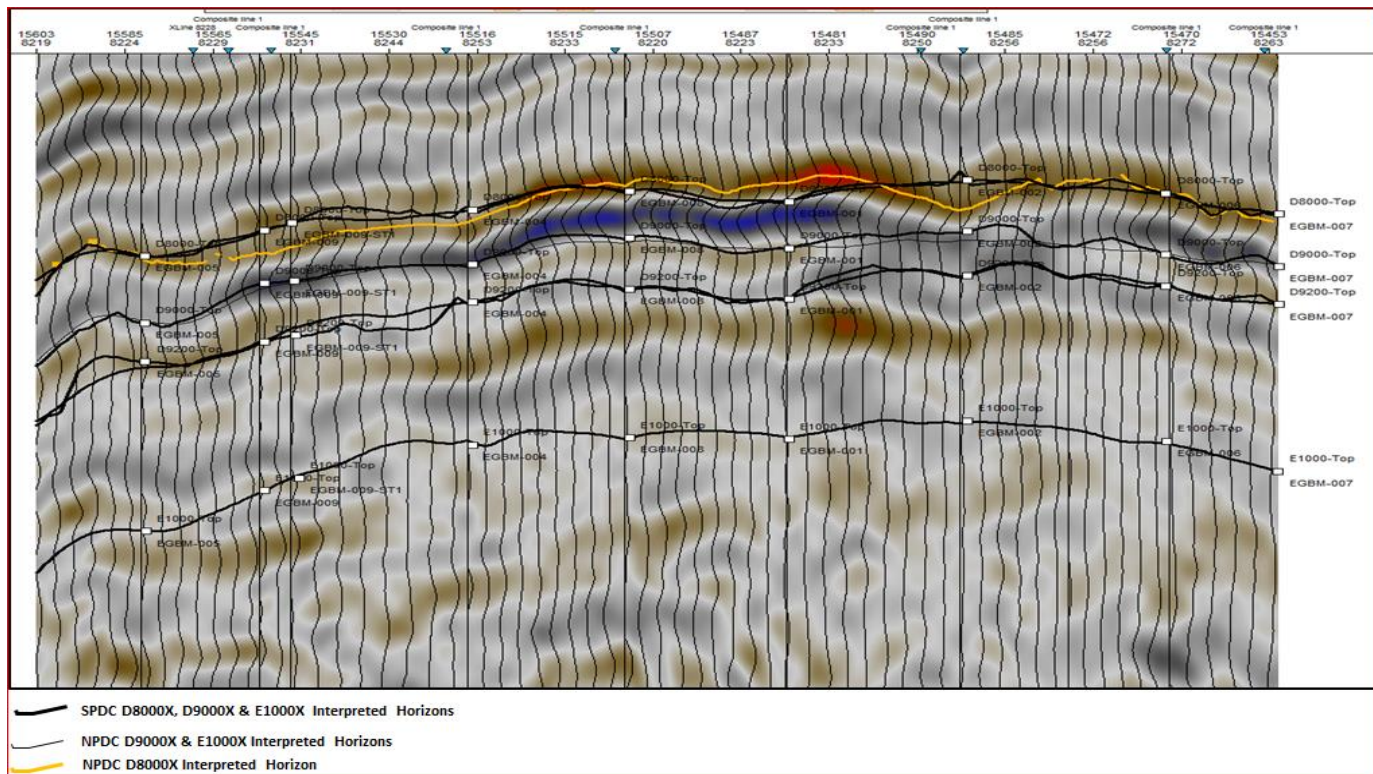


Figure 2-1: Difference between SPDC and NPDC interpreted D8000X and D9000X Horizons. The interpretation follows the seismic reflectivity, but it deviates, in some cases, when it is tied to well top as is the case for EGBM-01 and EGBM-02

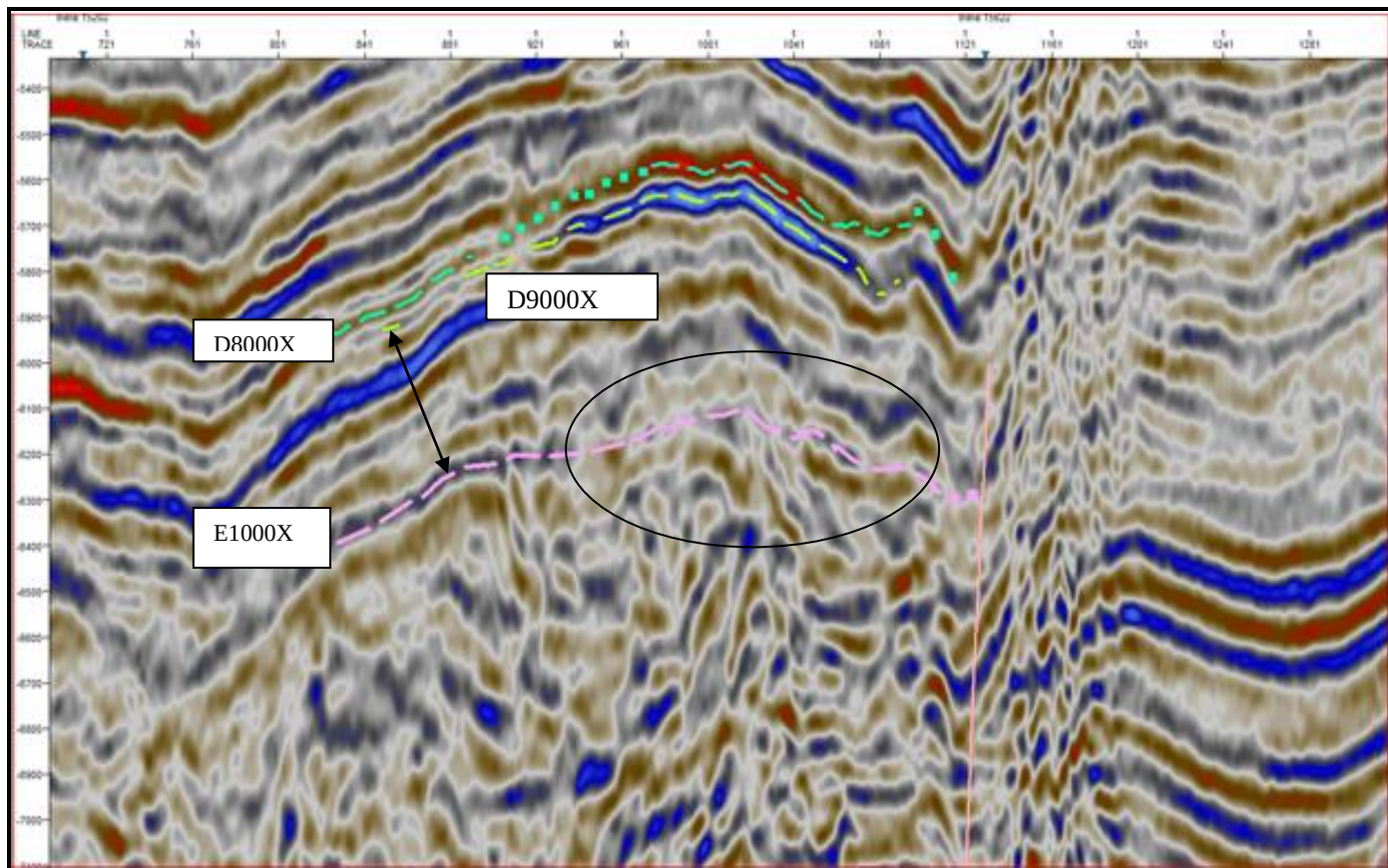


Figure 2-2: Egbema Main 2013 Reprocessed Seismic Volume (PSDM) in the depth domain showing D8000X, D9000X and E1000X Seismic. Circle shows poor lateral continuity for E1000X

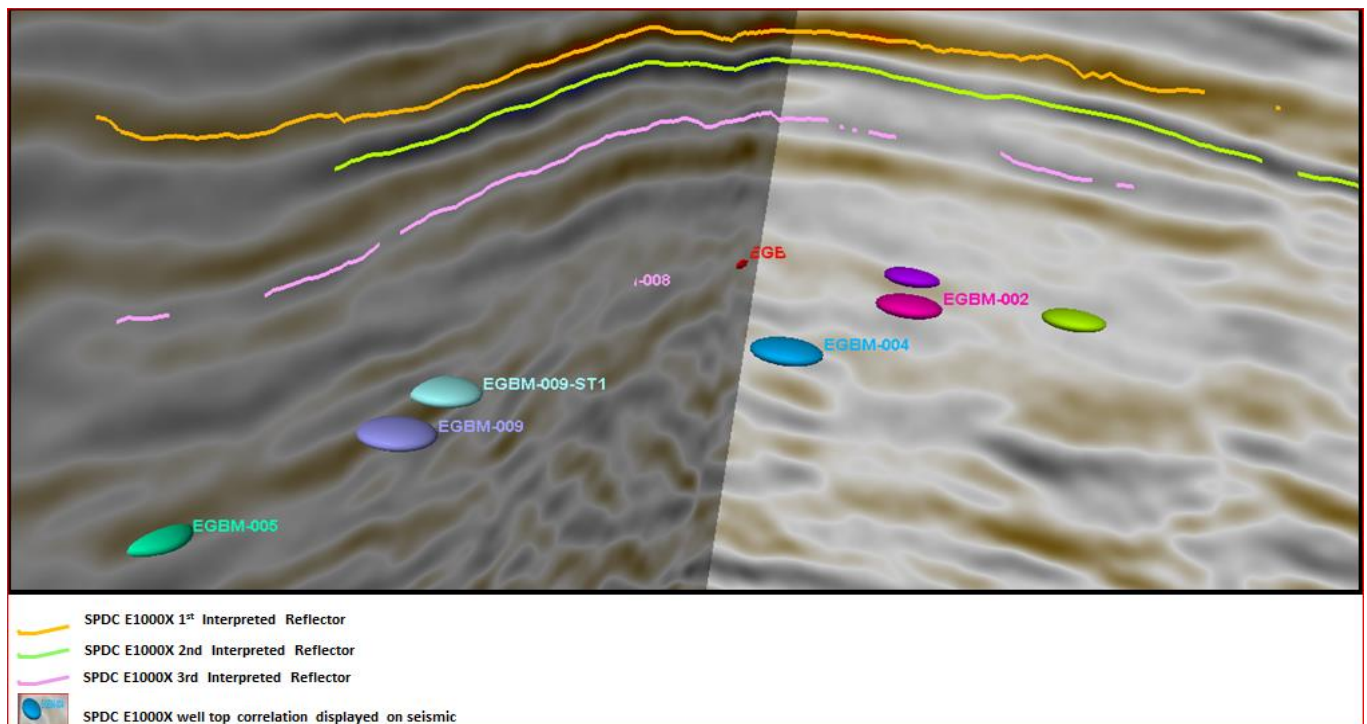


Figure 2-3: Three different seismic reflectors interpreted by SPDC for E1000 reservoir due to the difficulty in following one single reflector interpreted. Reflector three gave the smallest residual, hence accepted as the E1000 horizon.

Table 2-3: E1000X Depth Uncertainty Analysis

[illegible]

3. RESERVOIR GEOLOGY

3.1. Structural Geology

The Egbema field is on the same structural trend as the Egbema West and Ugada Fields (Figure 3-1). Egbema is bounded to the north and south by an east west trending synthetic growth fault that joined in the eastern part of the field. To the west it is separated from the Egbema-West Field by a saddle.

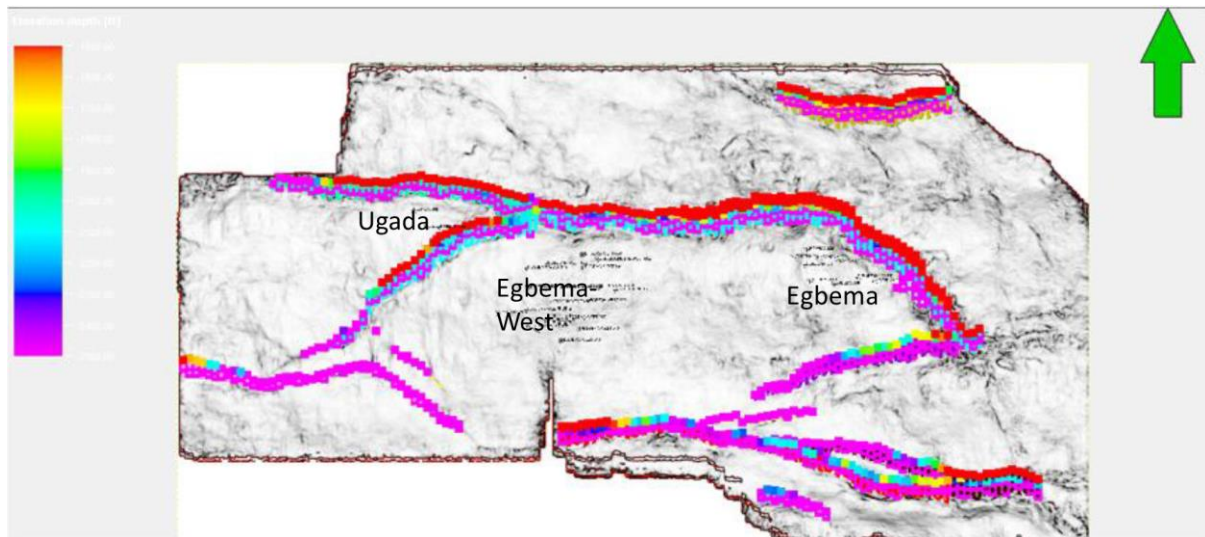


Figure 3-1:

3.2. Stratigraphy and Sedimentology

Reservoir Descriptions

The focus of this report is on three reservoirs in Egbema Field (D8000, D9000 & E1000). The reservoirs are all gas and oil bearing. The reservoir descriptions below are based on petrophysical well logs as no core was available for the interval of interest. All the sands are overlain and underlain by shale layers of variable thickness and the reservoirs are stacked upon each other with the shallowest being the D8000 and the deepest the E1000, with the D9000 in between them.

D8000

The D8000 sands are ~72ft thick and bounded by ~27ft shale and separated from the D9000 sands at the base by ~32ft of shale. The D8000 sands consist of coarsening upward sequence, capped by blocky layer. In some wells the entire D8000 are made of blocky layers of sands. The D8000 is interpreted to represent mainly shorface deposits, incised by channels. The D8000 sands are laterally extensive across the entire field and are correlatable. The basal unit of the reservoir are poorly developed.

D9000

The D9000 is interpreted as a complex and divided into four subunits. The D9000 sands are ~253ft thick and bounded by ~32ft shale separating it from the D8000 sand above and a ~110ft shale from the E1000 sand below. The D9000 sands is predominantly

channel sands with a sharp base and blocky log profile with shale intercalations. The D9000 sands are laterally extensive across the entire field and are correlatable.

E1000

The E1000 sands are ~67ft thick and bounded by ~110ft shale separating it from the D9000 sands above, a ~38ft shale at the base separate it from the E3000 sand. The E1000 sands consist of coarsening upward sequence, capped by blocky layer. The sands are highly serrated, which is an indication of tidal influence. Hence the interpretation of a tidal channel environment of deposition.

3.3. Conceptual Depositional Model

No detailed facies model was built as it was agreed to keep the model simple as this was for volume validation pre-DG1. However based on the log signatures for the wells penetrating the reservoirs the Egbema D8000, D9000 and E1000 are interpreted as shallow marine environment of depositions, with predominantly channels incision into shoreface environment.

3.4. Correlation

A field wide and reservoir wide correlation (for D8000, D9000 & E1000 sands) was carried out as a means of validating the reservoir tops and bases to ensure consistency of the reservoir picks, sand nomenclature and to identify correlatable intra reservoir units (Figure 3-2). The picks were found to be consistent in all wells.

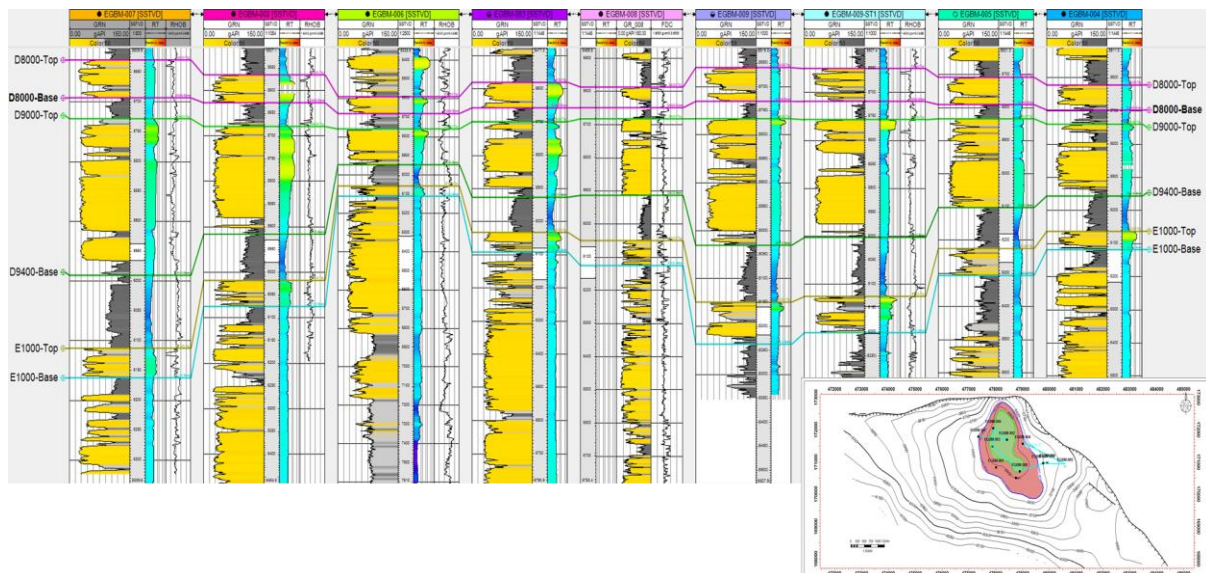


Figure 3-2: Field wide correlation of D8000, D9000 and E1000 reservoirs

4. ROCK AND FLUID PROPERTIES

4.1. Contacts

The three modelled reservoirs in Egbema composed of oil rim overlain by gas cap. The fluid contacts for the reservoirs tabulated below in Table 4-1 and Figures 4-1- 4-3. They have been derived mainly from the log data, sidewall samples and mudlogging data. No pressure data information was available to back up the derived contacts.

Table 4-1

Reservoir	GOC (ftss)	OWC (ftss)
D8000	5587	5630
D9000	5711	5803
E1000	6104	6174

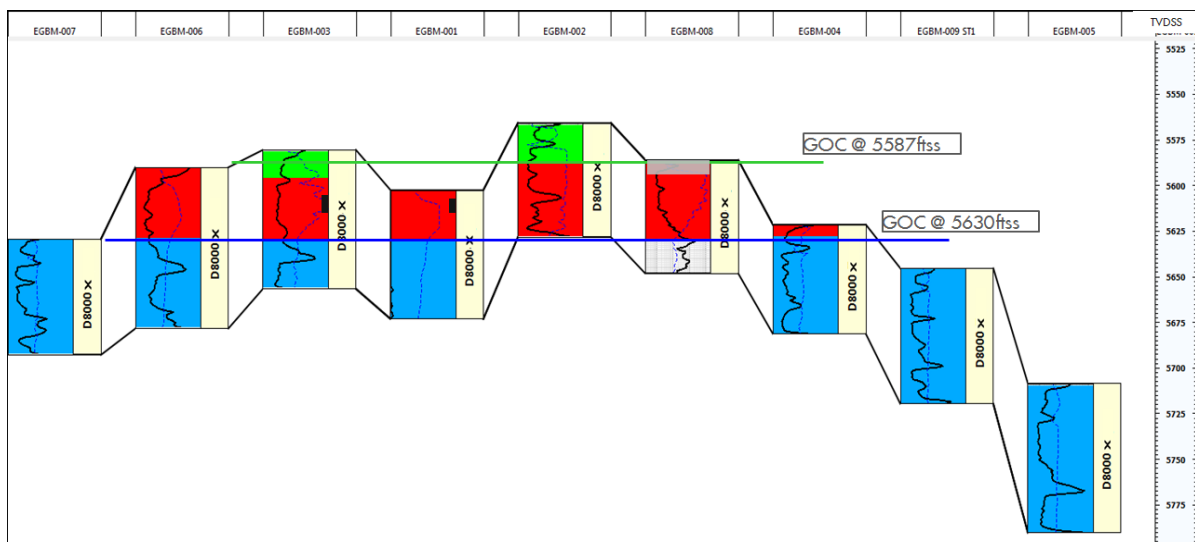


Figure 4-1: Hydrocarbon Distribution Plot for D8000 reservoir

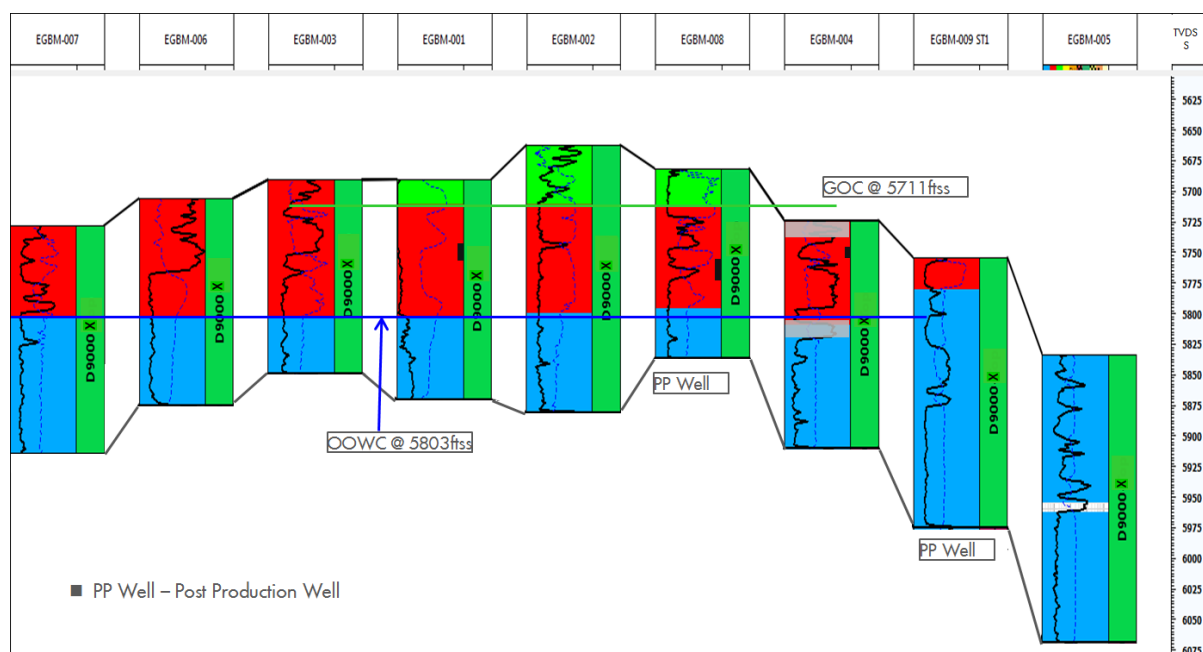


Figure 4-2: Hydrocarbon Distribution Plot for D9000 Reservoir

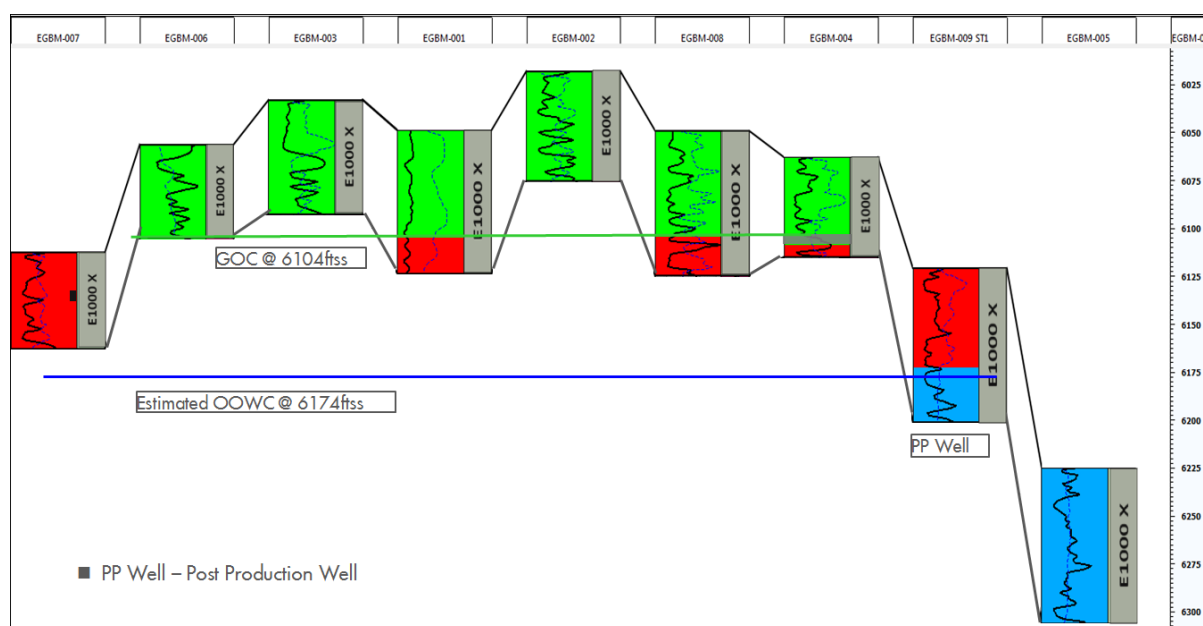


Figure 4-3: Hydrocarbon Distribution Plot for E1000 Reservoir

4.2. Rock Properties

A new petrophysical evaluation was carried out to validate NPDC reported petrophysical properties. The work scope covered the detailed review of sand tops/bases, porosity and saturation determination, fluid typing and contacts analyses.

Below is the summary of the sums and averages (Table 4-2) for pay zones of each well NtG, Porosity and Water Saturation computed and used in the modelling work. Details of the Petrophysical work will be documented in the associated Petrophysical report currently being put together by the petrophysicist.

Table 4-2: Egbema D8000, D9000 & E1000 Property ranges

Reservoir	Egbema Property Ranges											
	N/G			Por			Sg			So		
	P90	P50	P10	P90	P50	P10	P90	P50	P10	P90	P50	P10
D8000	0.81	0.86	0.90	0.27	0.29	0.30	0.47	0.66	0.84	0.63	0.76	0.89
D9000	0.63	0.72	0.80	0.27	0.29	0.31	0.55	0.62	0.70	0.46	0.57	0.67
E1000	0.69	0.76	0.82	0.24	0.26	0.28	0.59	0.70	0.81	0.50	0.64	0.77

4.3. PVT

A review of all available fluid properties data in Egbema field showed that fluid sample from only one reservoir (E1000) was collected and analysed in the laboratory. Four more laboratory analysis reports were however obtained from two nearby fields (Egbema West and Oguta).

In addition to the limited laboratory reports, reservoir solution gas-oil ratio were reasonably understood from early production data (the three reservoirs of interest have significant production histories) while the oil densities were known from historical produced oil analyses.

To assess oil formation volume factors the solution gas-oil ratios, oil densities, as well as reservoir temperatures and bubble point pressures (taken as reservoir initial pressures corrected to the gas oil contact depths) were used as inputs into two fluid properties correlations (Standing Correlation and the Niger Delta Correlation) to obtain the formation volume factors

A third set of evaluations were carried out by plotting the formation volume factors obtained from laboratory reports of fluid properties in Egbema and nearby fields against depths of reservoirs and solution gas-oil ratio (Rsi) given that Rsi were known to a reasonable certainty from early production producing gas oil ratio (Figure 4.4). The Boi vs. Rsi correlation was eventually used to determine Boi for target reservoirs given that correlation had a better fit with data than the Bi vs. depth plot as seen by their R² values.

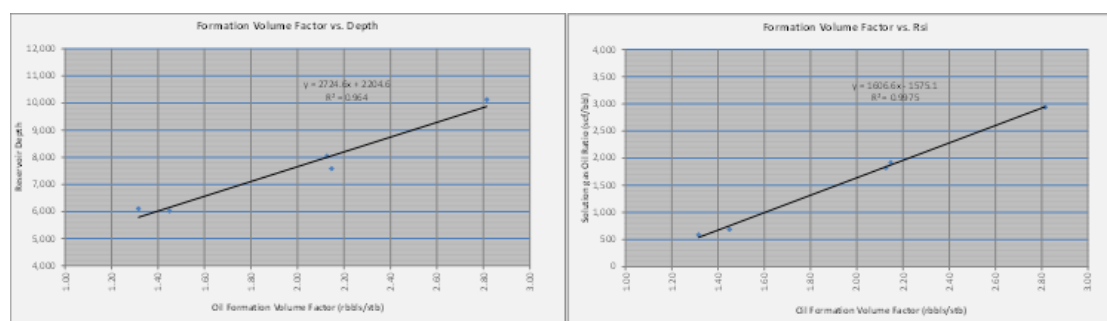


Figure 4-4: Egbema and Nearby Fields Formation Volume Factor Correlation with Depth and Solution Gas-Oil Ratio

The laboratory measured formation volume factor for the E1000 reservoir (1.3 rbbls/stb) compared well with the value obtained from the three methods (1.28 – 1.3) rbbls/stb. Table 4.2 shows the Boi values from the three methods described above.

Table 4-3: Formation Volume Factor Obtained From Three Methods

Field	Reservoir	FVF (rbbls/stb)		
		Field Corelation	Standing	Niger Delta
EGBEMA	D8000X	1.129	1.162	1.112
	D9000X	1.200	1.192	1.141
	E1000X	1.283	1.290	1.296

Table 4-4: Oil Formation Volume Factor (Boi) Uncertainty Spread.

Field	Reservoir	FVF (rbbls/stb)		
		Low	Mid	High
EGBEMA	D8000X	1.162	1.129	1.112
	D9000X	1.200	1.192	1.141
	E1000X	1.296	1.290	1.283

Gas formation volume factors (Bgi) were simply obtained from the Standing correlation (Figure 4.5), given that the Nigeria correlation does not generate gas properties and no reports of laboratory analysed gas samples were obtained. Table 4.4 summarises the Bgi values evaluated.

Gas correlations (Standing)	
YN2	0
YCO2	0.006
YH2S	0
Z	0.887665213
E	160.1404832
Ugas	1.6586E-05 Pa.s 0.0165862 cP

Figure 4-5: Formation Volume Factor Evaluation for Egbema D8000X Using Nigeria Correlation**Table 4-5: Gas Formation Volume Factors**

Field	Reservoir	FVF (cf/scf)
Egbema	D8000X	0.006289
	D9000X	0.006230
	E1000X	0.00584

5. STATIC RESERVOIR MODELLING

5.1. Structural Modelling

Straightforward grid structure was used in the modelling of Egbema Field due to lack of intra-reservoir faults. Pillar gridding was used to define the model boundary, with cells aligned along the field axis. A 100m x 100m grid was used based on considerations of the spatial extent of the oil rim and well spacing to capture the smallest representative elementary volumes that was considered important for this modelling exercise bearing in mind that this is pre-DG1, hence the basis for the coarse model.

Layering was done considering the entire reservoir unit as a single layer, because detailed facies modelling was not necessary at this level and average reservoir properties was been considered for the property modelling workflow.

Given the lack of faults, horizons were inputted as surfaces and tied to the wells during the Make Horizon process, no radius of influence was defined. The base horizons were derived from isochore and made conformable to the top structure in the Make Zone process.

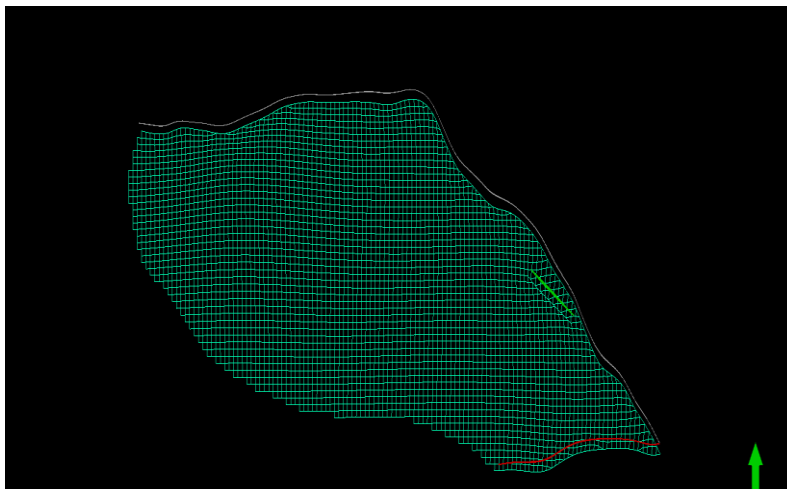


Figure 5-1: View of the modelled AOI for the Egbema Field

5.2. Property Modelling

At this stage no detailed property modelling was performed; simple sums and average properties were used to populate the model. The model was designed to be kept simple as no detailed facies model was done. Sums and averages were defined for porosity, NtG and Hydrocarbon Saturation as discussed in section 4.2. As the primary purpose of this modelling work was for volumetric purposes, this simple sums and average approach suffices.

6. UNCERTAINTY

Structure

Three structural realisations were derived for the three reservoirs based on the depth uncertainty discussed in section 2.1.

Properties

Following the structural realisations built, the next steps are straightforward, with the model layered and then values for NtG, Porosity, Hydrocarbon saturation calculated with an individual uncertainty range. Ranges are captured in Tables 6-1 - 6-3 below.

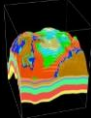
Table 6-1: Egbema D8000 Uncertainty Matrix

		Low	Base	High
	Structure	-52ft. Depth uncertainty	Base case top structure	+52ft. Depth uncertainty
	Porosity	0.27	0.29	0.3
	Saturation	Oil - 0.63 Gas - 0.47	Oil - 0.76 Gas - 0.66	Oil - 0.89 Gas - 0.84
	Fluid Prop.		Oil - 1.129 (rbbls/stb) Gas - 0.006289 (cf/scf)	
	N/G	0.81	0.86	0.9

Table 6-2: Egbema D9000 Uncertainty Matrix

		Low	Base	High
	Structure	-41ft. Depth uncertainty	Base case top structure	+41ft. Depth uncertainty
	Porosity	0.27	0.29	0.31
	Saturation	Oil - 0.46 Gas - 0.55	Oil - 0.57 Gas - 0.62	Oil - 0.67 Gas - 0.70
	Fluid Prop.		Oil - 1.192 (rbbls/stb) Gas - 0.006230 (cf/scf)	
	N/G	0.63	0.72	0.8

Table 6-3: Egbema E1000 Uncertainty Matrix

		Low	Base	High
	Structure	-78ft. Depth uncertainty	Base case top structure	+78ft. Depth uncertainty
	Porosity	0.24	0.26	0.28
	Saturation	Oil - 0.50 Gas - 0.59	Oil - 0.64 Gas - 0.70	Oil - 0.77 Gas - 0.81
	Fluid Prop.		Oil - 1.290 (rbbls/stb) Gas - 0.00584 (cf/scf)	
	N/G	0.69	0.76	0.82

7. VOLUMETRIC SUMMARY

Volumes are calculated using the Petrel model and recorded below in Table 7-1 & 7-2.

Table 7-1: Egbema D8000, D9000 & E1000 base case static model volumes (STOIIP)

Reservoir	GRV (acre-ft)	NRV (acre-ft)	NPV (acre-ft)	HCPV (acre-ft)	Boi (rb/stb)	STOIIP (MMstb)
D8000	28413.87	24435.93	7086.42	5385.68	1.129	37.01
D9000	127298.87	91655.19	26580.01	15150.60	1.192	98.61
E1000	68247.02	51867.73	13485.61	8630.79	1.290	51.91

Table 7-2: Egbema D8000, D9000 & E1000 base case static model volumes (GIIP)

Reservoir	GRV (acre-ft)	NRV (acre-ft)	NPV (acre-ft)	HCPV (acre-ft)	Bgi (ft ³ /ft ³)	GIIP (Bscf)
D8000	3600.52	3096.44	897.97	592.66	0.006289	4.10
D9000	16306.63	11740.78	3404.82	2110.99	0.006233	14.75
E1000	41629.86	31638.69	8226.06	5758.24	0.00584	42.95

Uncertainty ranges in the reservoirs of interest are given in Tables 7-3, reflecting both uncertainties for STOIIP and GIIP individually.

Table 7-3: Egbema D8000, D9000 & E1000 Volume Ranges

Reservoir	STOIIP (MMstb)			GIIP (Bcf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000	24.09	37.01	55.21	1.99	4.10	7.94
D9000	82.93	98.61	119.98	9.8	14.75	21.39
E1000	36.06	51.91	68.97	33.08	42.95	67.03

The relative impact of the various parameters is shown for the STOIIP and GIIP in figures 7-1, 7-2 and 7-3 for the D8000, D9000 and E1000 respectively.

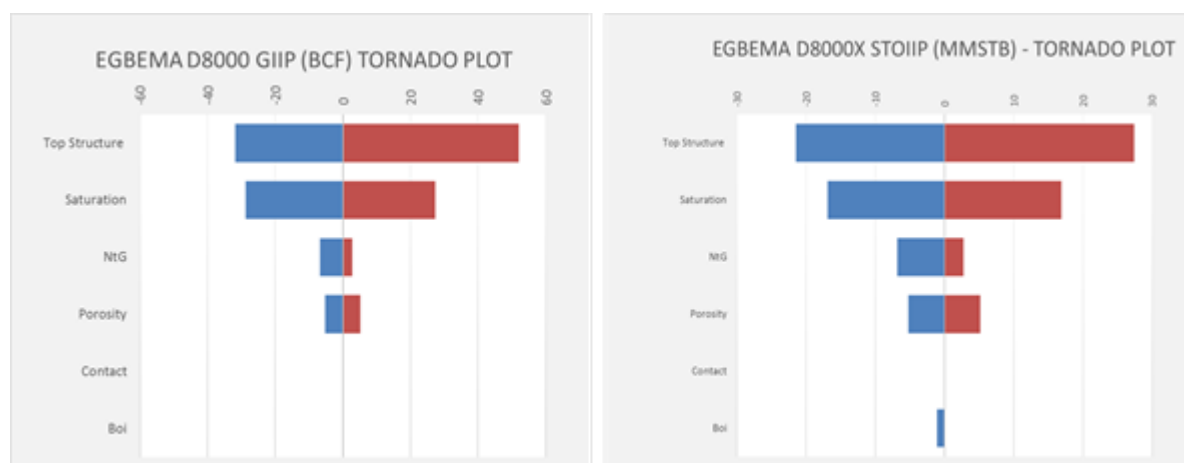


Figure 7-1: Tornado plot for key variable for D8000 STOIIP and GIIP showing the influence of Top structure and saturations on volumes

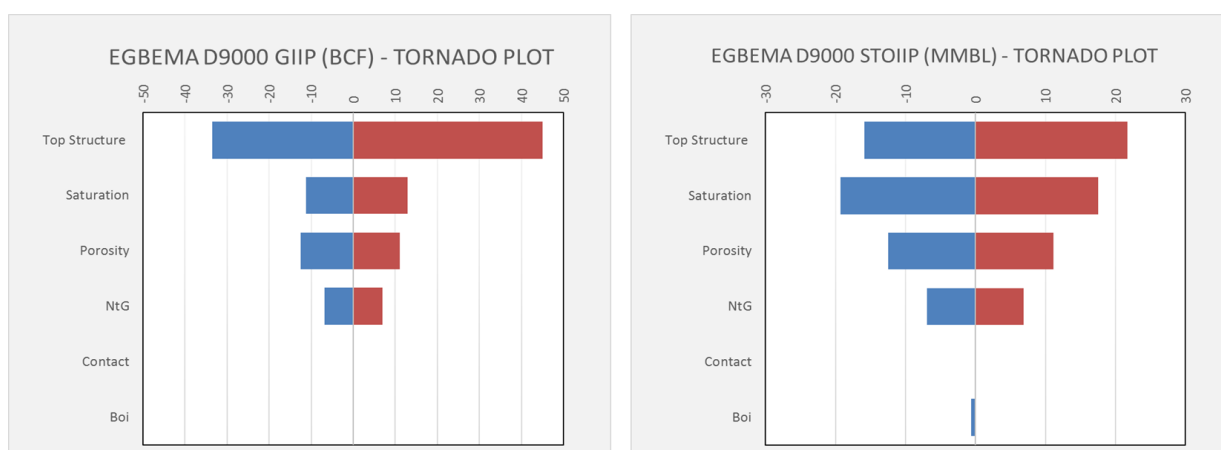


Figure 7-2: Tornado plot for key variable for D9000 STOIP and GIIP showing the influence of Top structure and saturations on volumes

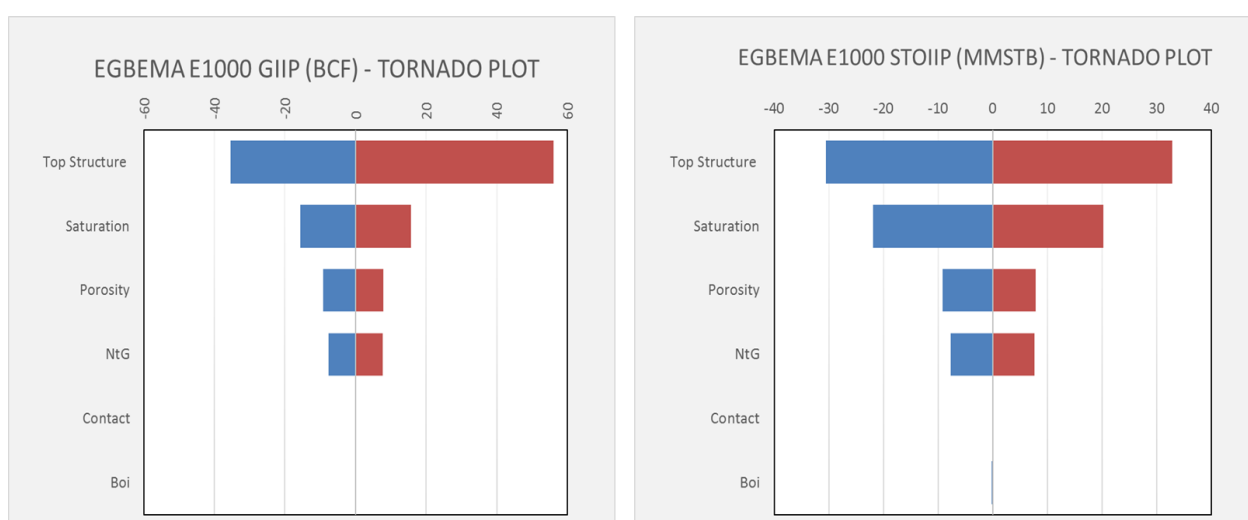


Figure 7-3: Tornado plot for key variable for D8000 STOIP and GIIP showing the influence of Top structure and saturations on volumes

Volume sensitivities show that “Structure and Hydrocarbon Saturation” are the most impacting parameters on STOIP for three reservoirs, while Structure is the most impacting for GIIP except for D8000 where Saturation also has a major impact.

The outcome of these sensitivities helped in the stringing of the parameters in coming up with the P10 and P90 volumes as documented in Table 7-3. Tables 7-4 – 7-6 shows the stringing of the STOIP and GIIP input parameters

Table 7-4: D8000 Stringing of Parameters

Egbema D8000X Reservoir in Place (GIIP) Volume Realisations					
Uncertainty	Structure (GRV)	N/G	Porosity	Saturation	Bgi
LOW					
LOW	LC=2456.53	LC = 0.81	LC = 0.27	LC = 0.47	Bgi = 0.006289
BASE	BC = 3600.52	BC = 0.86	BC = 0.29	BC = 0.66	Bgi = 0.006289
BASE					
HIGH	HC = 5473.36	HC = 0.90	HC = 0.30	HC = 0.84	Bgi = 0.006289
HIGH					

Egbema D8000X Reservoir in Place (STOIIP) Volume Realisations					
Uncertainty	Structure (GRV)	N/G	Porosity	Saturation	Boi
LOW					
LOW	LC=22311.86	LC = 0.81	LC = 0.27	LC = 0.63	Boi = 1.182
BASE	BC = 28413.86	BC = 0.86	BC = 0.29	BC = 0.76	Boi = 1.129
BASE					
HIGH	HC = 36194.52	HC = 0.90	HC = 0.30	HC = 0.89	Boi = 1.112
HIGH					

Table 7-5: D9000 Stringing of Parameters

Egbema D9000X Reservoir in Place (GIIP) Volume Realisations					
Uncertainty	Structure (GRV)	N/G	Porosity	Saturation	Bgi
LOW					
LOW	LC=10837.27	LC = 0.63	LC = 0.27	LC = 0.55	Bgi = 0.006230
BASE	BC = 16306.63	BC = 0.72	BC = 0.29	BC = 0.62	Bgi = 0.006230
BASE					
HIGH	HC = 22644.85	HC = 0.80	HC = 0.31	HC = 0.70	Bgi = 0.006230
HIGH					

Egbema D9000X Reservoir in Place (STOIIP) Volume Realisations					
Uncertainty	Structure (GRV)	N/G	Porosity	Saturation	Boi
LOW					
LOW	LC=107051.86	LC = 0.63	LC = 0.27	LC = 0.46	Boi = 1.200
BASE	BC = 127298.87	BC = 0.72	BC = 0.29	BC = 0.57	Boi = 1.192
BASE					
HIGH	HC = 154980.51	HC = 0.80	HC = 0.31	HC = 0.67	Boi = 1.141
HIGH					

Table 7-6: E1000 Stringing of Parameters

Egbema E1000X Reservoir in Place (GIIP) Volume Realisations					
Uncertainty	Structure (GRV)	N/G	Porosity	Saturation	Bgi
LOW					
LOW	LC=32058.17	LC = 0.69	LC = 0.24	LC = 0.59	Bgi = 0.00584
BASE	BC = 41629.86	BC = 0.76	BC = 0.26	BC = 0.70	Bgi = 0.00584
BASE					
HIGH	HC = 64967.68	HC = 0.82	HC = 0.28	HC = 0.81	Bgi = 0.00584
HIGH					

Egbema E1000X Reservoir in Place (STOIIP) Volume Realisations					
Uncertainty	Structure (GRV)	N/G	Porosity	Saturation	Boi
LOW					
LOW	LC=47416.03	LC = 0.69	LC = 0.24	LC = 0.50	Boi = 1.296
BASE	BC = 68247.02	BC = 0.76	BC = 0.26	BC = 0.64	Boi = 1.280
BASE					
HIGH	HC = 90682.92	HC = 0.82	HC = 0.28	HC = 0.77	Boi = 1.283
HIGH					

A summary of the base case volumetrics differences between this study and the NNS ARPR 2017 is shown in Table 7-7 below.

Table 7-7: Base case STOIP and GIIP comparison Static Model vs ARPR 2017

	Current Studies		ARPR (01.01.2017)		Variation to ARPR 2017 (%)	
	STOIP	GIIP	STOIP	GIIP	STOIP	GIIP
D8000	37.01	4.10	37.01	4.10	0	0
D9000	98.61	14.75	84.80	16.70	-16	-12
E1000	51.91	42.95	60.80	40.20	15	-7

8. CONCLUSION & RECOMMENDATION

This report summarized the work done to shadow the Egbema FOD studies by NPDC. It was agreed in house to keep the model simple since it was a pre-DG1 work.

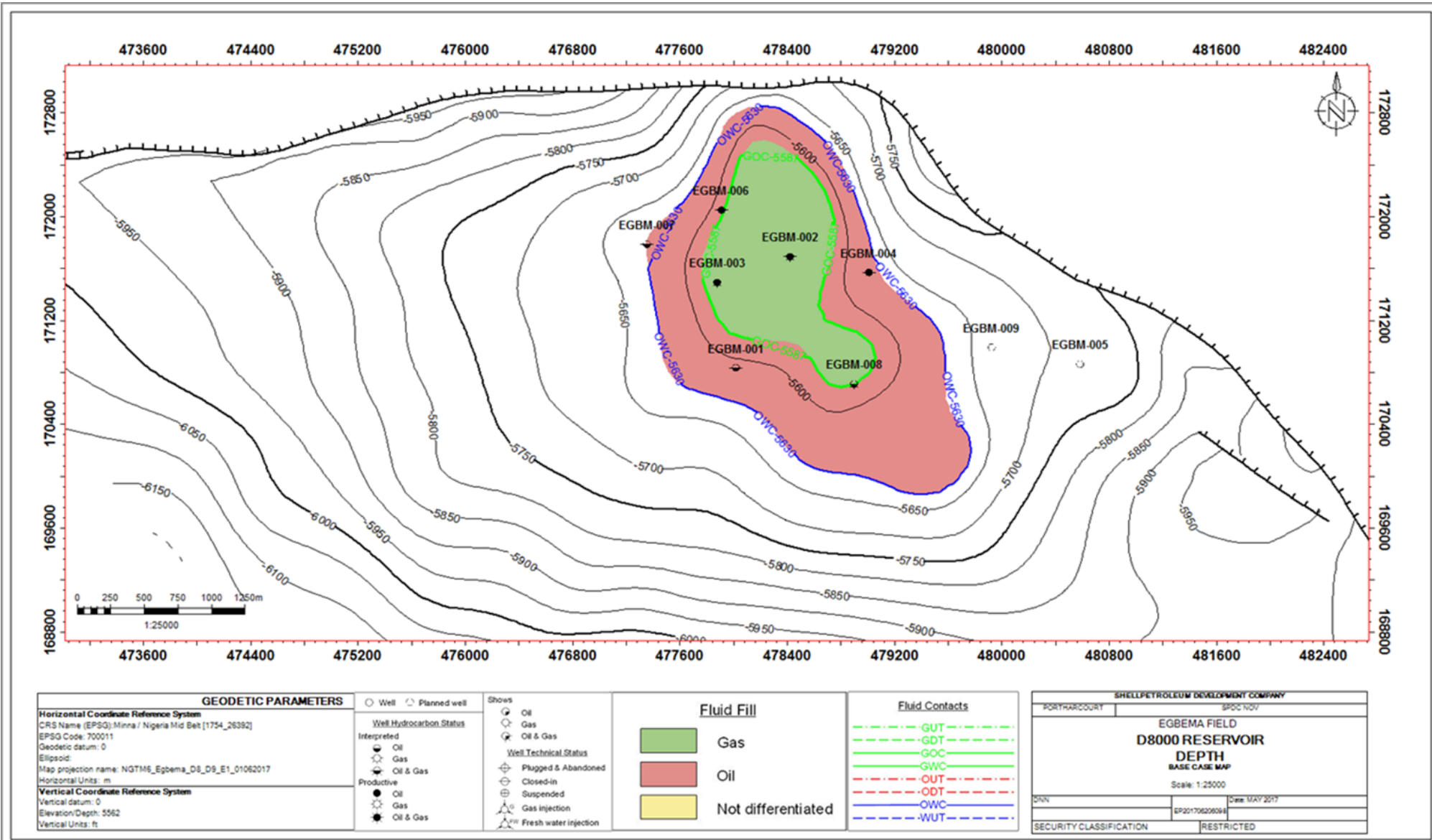
As part of future phases of work (Pre-DG2) it is recommended to look in details:

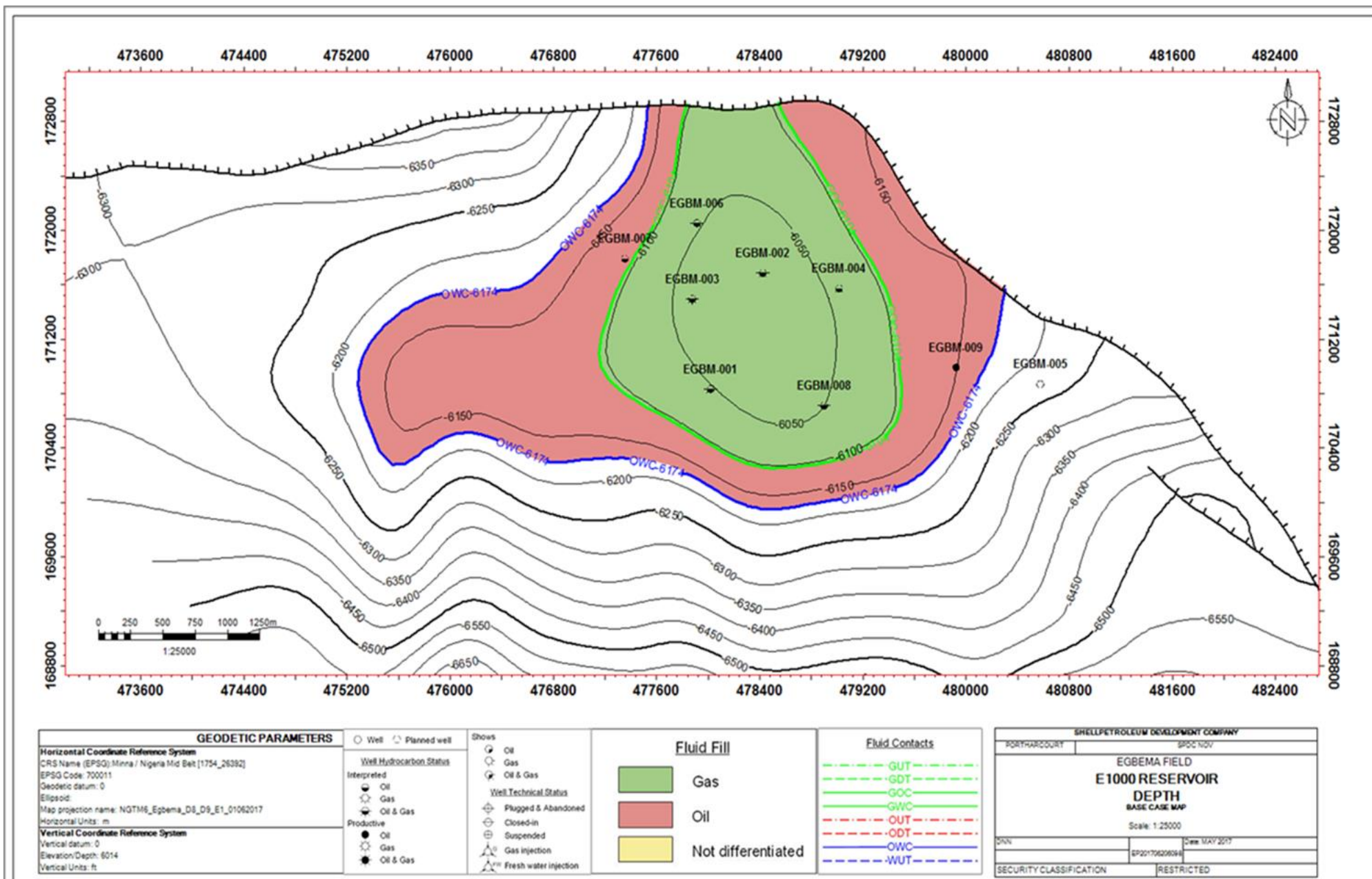
1. Conceptual depositional environment and detailed facies definition.
2. Modelling of well data and not simply well averages should be performed to capture and possible heterogeneities.

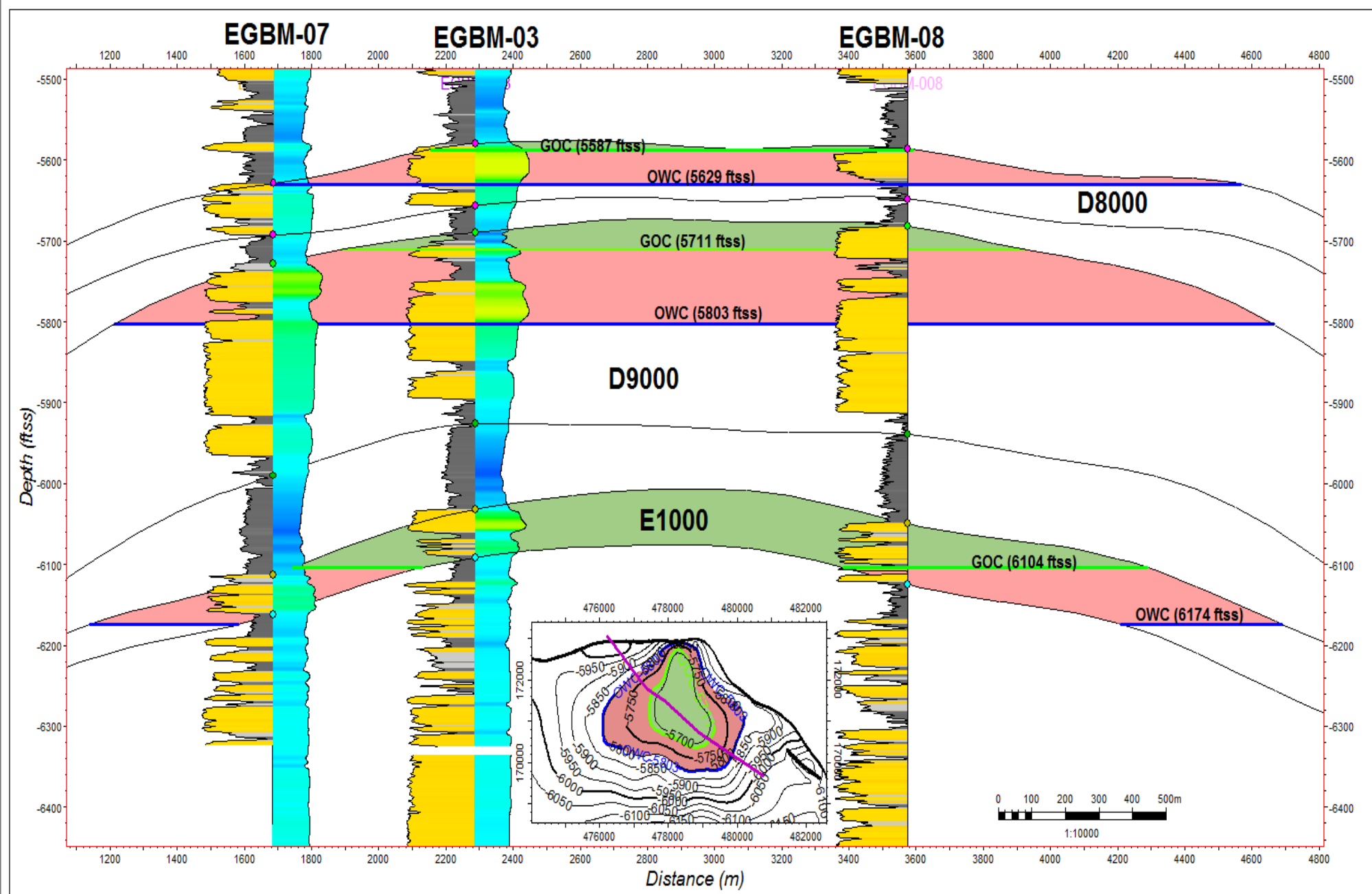
REFERENCES

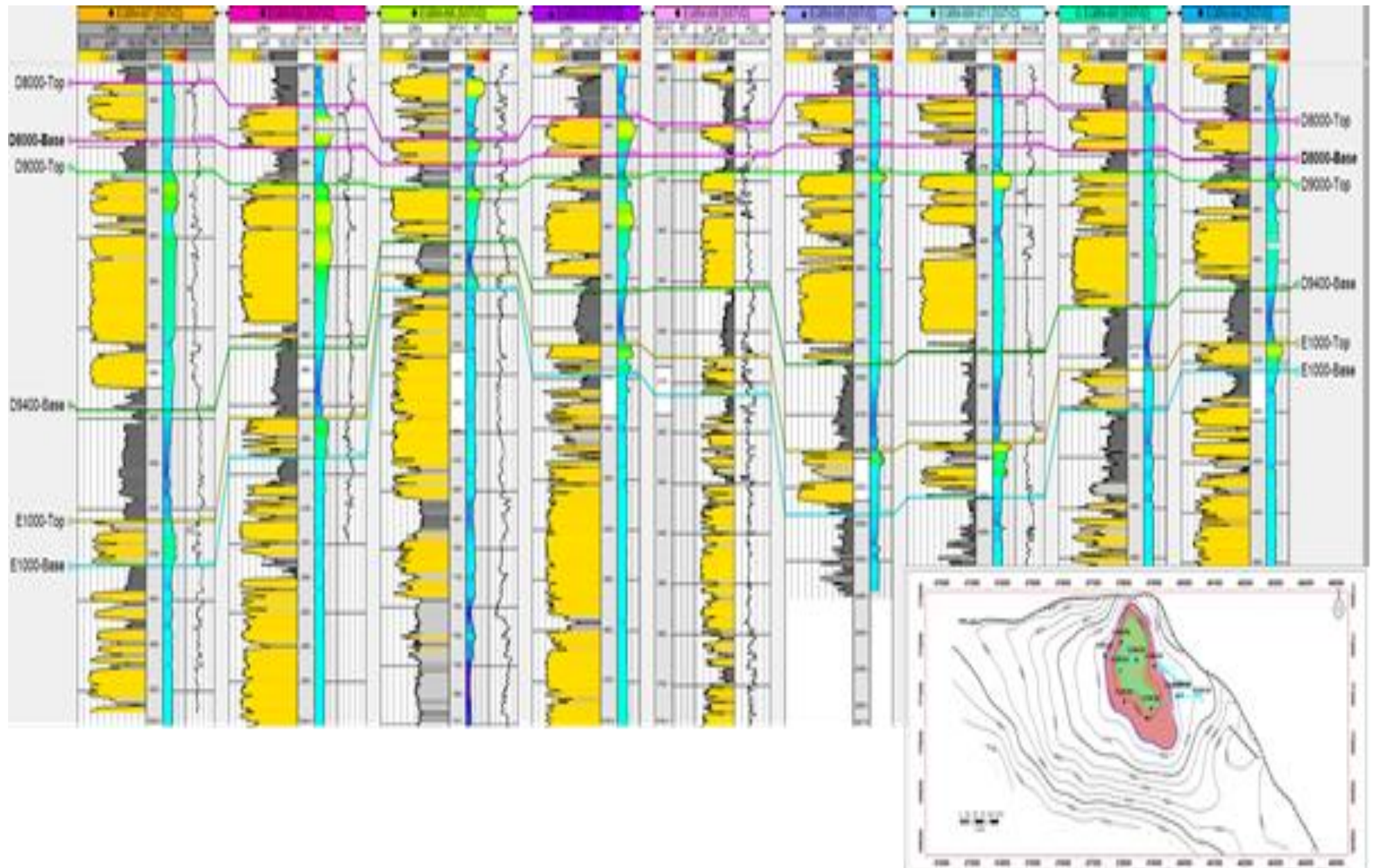
1. Study of Egbema Field , Nigeria, Part 1 Production Geology, by PDD Shell International Petroleum Maatschappij B.V., April 1988 (EP 58866/1)

APPENDIX 1: EGBEMA D8000, D9000 & E1000 TOP STRUCTURE MAPS, CROSS SECTION & CORRELATION PANEL









APPENDIX 2: EGBEMA D8000, D9000 & E1000 SAND TOPS AND BASES UPDATE

Reservoir	Well	Well Tops (MD)			Well Tops (ftss)			Well Bases (MD)			Well Bases (ftss)		
		Current	CDS	Diff	Current	CDS	Diff	Current	CDS	Diff	Current	CDS	Diff
D8000	Egbema-001	5685.21	5690.07	-4.86	5606	5611	-5	5757.83	5757.96	-0.13	5678	5679	-1
D8000	Egbema-002	5639.00	5639.00	0.00	5565	5565	0	5700.42	5701.60	-1.18	5627	5628	-1
D8000	Egbema-003	5654.24	5654.24	0.00	5580	5580	0	5731.30	5730.37	0.93	5657	5656	1
D8000	Egbema-004	5696.03	5696.03	0.00	5621	5621	0	5771.79	5755.57	16.22	5697	5681	16
D8000	Egbema-005	5786.27	5786.27	0.00	5708	5708	0	5868.88	5868.24	0.64	5791	5790	1
D8000	Egbema-006	5664.21	5670.03	-5.82	5590	5596	-6	5752.53	5752.53	0.00	5678	5678	0
D8000	Egbema-007	5705.27	5705.27	0.00	5628	5629	-1	5768.74	5777.20	-8.46	5693	5701	-8
D8000	Egbema-008	5669.38	5673.68	-4.30	5586	5591	-5	5731.59	5742.42	-10.83	5649	5659	-10
D8000	Egbema-009	5751.56	5590.26	161.30	5659	5498	161	5821.99	5687.10	134.89	5730	5595	135
D8000	Egbema-009st	5827.97	5823.44	4.53	5650	5645	5	5901.74	5903.38	-1.64	5718	5720	-2
D9000	Egbema-001	5772.92	5779.16	-6.24	5694	5700	-6	6039.70	6034.42	5.28	5960	5955	5
D9000	Egbema-002	5752.07	5752.25	-0.18	5662	5678	-16	5989.80	5978.92	10.88	5916	5905	11
D9000	Egbema-003	5773.63	5779.76	-6.13	5690	5705	-15	6000.90	5972.11	28.79	5926	5897	29
D9000	Egbema-004	5815.96	5816.63	-0.67	5724	5742	-18	6031.01	6046.76	-15.75	5956	5972	-16
D9000	Egbema-005	5911.67	5909.76	1.91	5834	5832	2	6180.19	6183.48	-3.29	6101	6105	-4
D9000	Egbema-006	5771.65	5780.53	-8.88	5706	5706	0	6020.02	6018.76	1.26	5946	5944	2
D9000	Egbema-007	5803.77	5814.63	-10.86	5728	5739	-11	6067.00	6041.88	25.12	6000	5966	34
D9000	Egbema-008	5765.00	5763.43	1.57	5682	5680	2	5995.88	5995.88	0.00	5913	5913	0
D9000	Egbema-009	5858.40	5728.78	129.62	5766	5637	129	6122.57	5821.94	300.63	6030	5730	300
D9000	Egbema-009st	5940.63	5944.23	-3.60	5754	5758	-4	6207.60		6207.60	6002		6002
E1000	Egbema-001	6131.63	6131.63	0.00	6052	6052	0	6206.42	6206.42	0.00	6127	6127	0
E1000	Egbema-002	6091.86	6091.86	0.00	6021	6018	3	6149.25	6149.25	0.00	6075	6075	0
E1000	Egbema-003	6106.71	6106.71	0.00	6032	6032	0	6166.28	6166.28	0.00	6092	6092	0
E1000	Egbema-004	6137.77	6137.77	0.00	6062	6063	-1	6189.83	6189.83	0.00	6115	6115	0
E1000	Egbema-005	6304.28	6304.28	0.00	6225	6225	0	6385.85	6385.85	0.00	6306	6306	0
E1000	Egbema-006	6130.54	6130.54	0.00	6056	6056	0	6184.1	6184.10	0.00	6110	6110	0
E1000	Egbema-007	6188.65	6188.65	0.00	6112	6112	0	6238.83	6238.83	0.00	6163	6163	0
E1000	Egbema-008	6132.38	6132.38	0.00	6049	6049	0	6207.95	6207.95	0.00	6125	6125	0
E1000	Egbema-009	6241.11	6242.39	-1.28	6149	6150	-1	6328.94	6314.50	14.44	6237	6222	15
E1000	Egbema-009st	6340.04	6338.36	1.68	6125	6124	1	6421.05	6421.41	-0.36	6201	6201	0